

Econ 2 - Lecture 7 - 1/28/26

Midterm Exam on Monday, February 2nd

Chapters 1, 2, 3

Head TA will send out a seating chart on Canvas

BRING A CALCULATOR AND PEN/PENCIL!

Equations given for: Growth Rate, CPI, Weighted Basket Cost, GDP Deflator, Unemployment Rate

Today: Finish up Full Employment (Chapter 3.3)

Last Class: Categories of Unemployment

Frictional + Seasonal + Structural

Long-term
but not
indicative
of a poor
economy

Naturally Occurring

Cyclical: Unemployment that is caused by direct changes in output

Example: US Car Industry in 2009

High gas prices + housing crash $\Rightarrow$ Lower Demand for Gas-Heavy Cars like Trucks & SUVs

Permanent Job Loss

Cyclical Unemployment $\Rightarrow$ changes in production (Y)

Full Employment Goal?

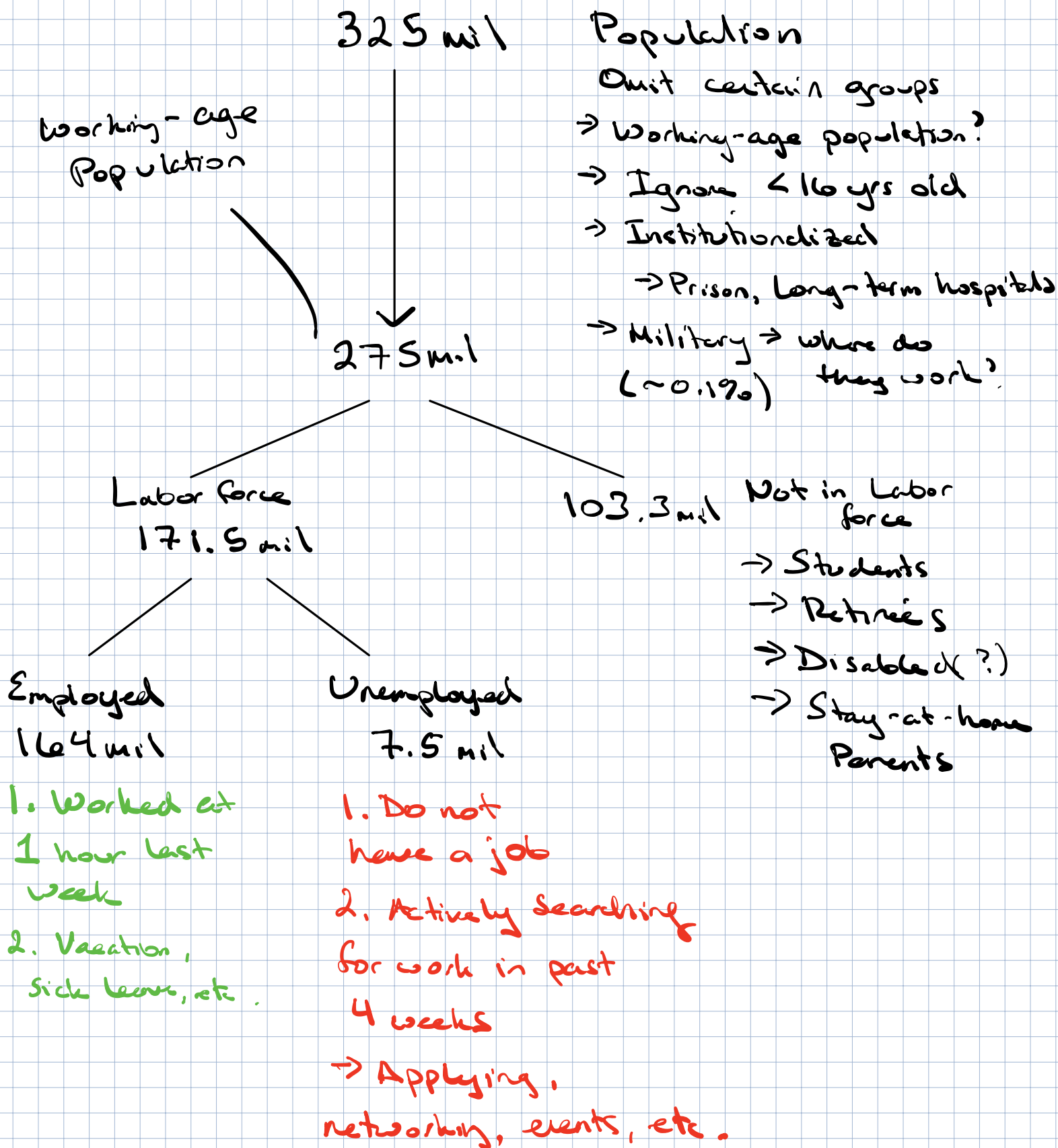
Cyclical Unemployment = 0%

$\hookrightarrow$ Output (Y) is at the full-employment level of production

$\hookrightarrow$ Full-Employment Output = $\bar{Y}$ (Y -bar)

Calculate the Unemployment Rate

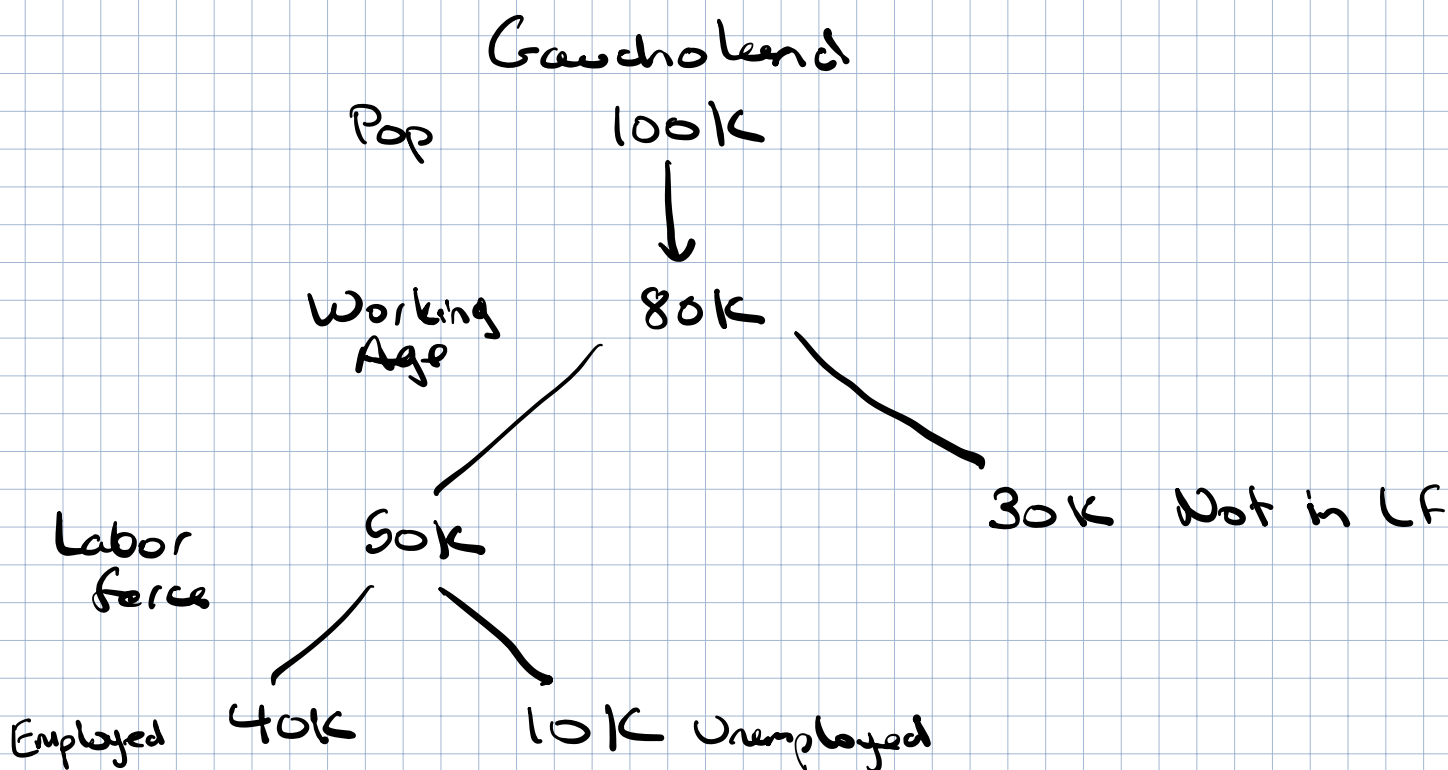
→ Bureau of Labor Statistics



What fraction of labor market is unemployed?

$$\text{BLS Unemployment Rate} = \frac{\text{Number of UE}}{\text{Labor force}} \times 100$$

$$\text{December 2025 UE} = \frac{7.5 \text{ mil}}{171.5 \text{ mil}} \times 100 = 4.40\%$$



$$\text{UE Rate} = \frac{10K}{50K} \times 100 = 20\%$$

$$\text{Labor Force Participation Rate} = \frac{\text{Labor force}}{\text{Working-Age Pop}} \times 100$$

Labor force = # of Employed + # of Unemployed

$$\text{LFP Rate} = \frac{40 + 10}{80} \times 100 = \frac{50}{80} \times 100 = 62.5\%$$

Shortcomings of Official BLS UE Rate

1. Individuals who work 1 hour = fully employed

Do you want to work more hours?

Are you involuntary part-time (IPT) worker?

Marco works 40 hours/week $\Rightarrow$ fully employed

Steve works 20 hours/week, wants 40 hours/week

Kelsey works 20 hrs/week, wants 40 hrs/week

Marco = 1 employed worker, 0 unemployed

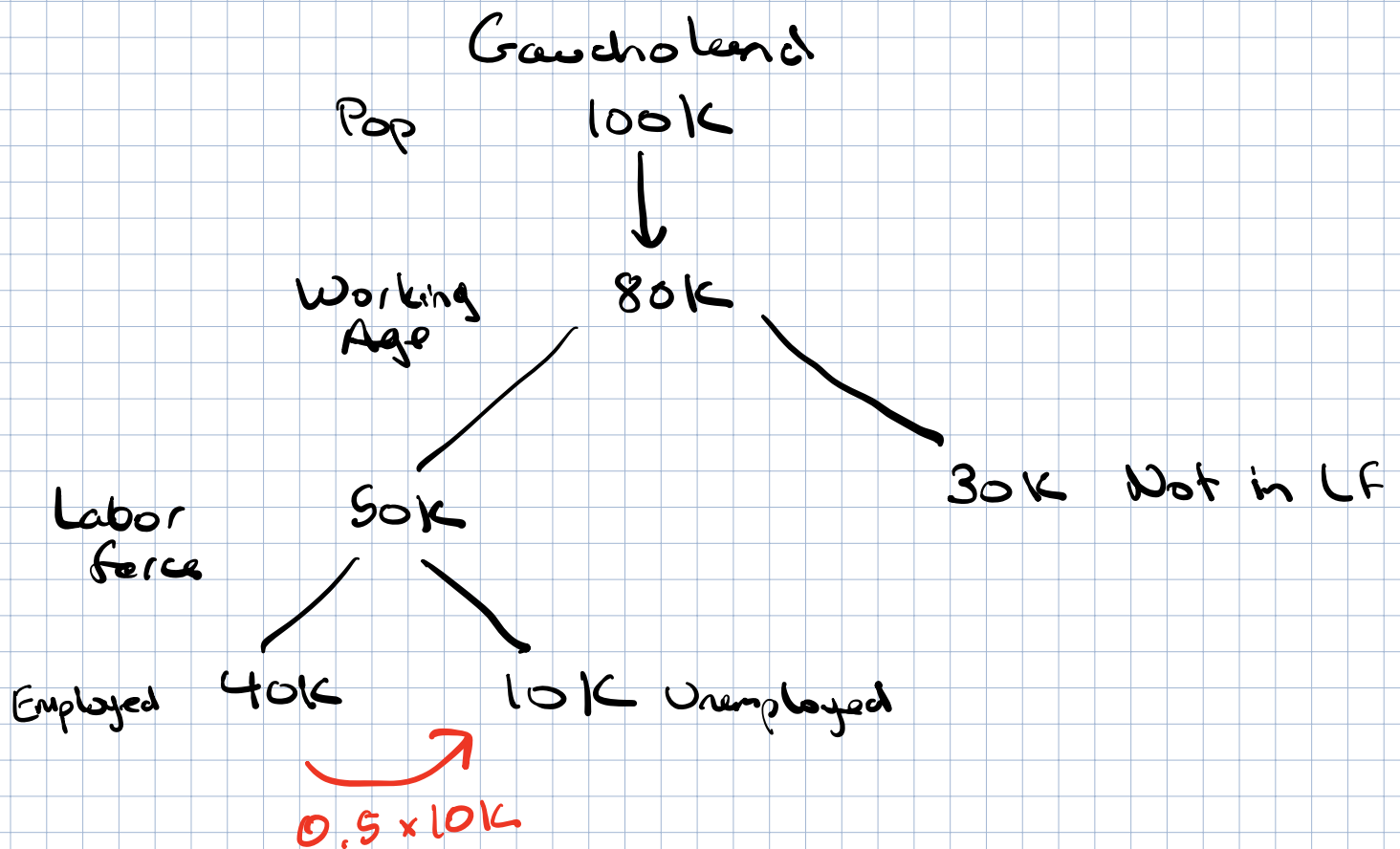
Steve & Kelsey = $\frac{1}{2}$ employed, $\frac{1}{2}$ unemployed

Want 40 hrs total (20 each)

$\Rightarrow$ Equivalent 1 unemployed worker

UE Rate
adjusted
for IPT
workers

$$\begin{aligned} &= \frac{\text{Unemployed} + 0.5 \times \text{IPT}}{\text{Labor force}} \times 100 \\ &= \frac{7.5 \text{ mil} + (0.5 \times 5.3 \text{ mil})}{171.5 \text{ mil}} \times 100 \\ &= \frac{10.15}{171.5} \times 100 = 5.9\% \end{aligned}$$



10K IPT $\rightarrow$ adjusted UE Rate?

$$\begin{aligned} \text{UE w/ IPT} &= \frac{\text{UE} + 0.5 \times \text{IPT}}{\text{LF}} \times 100 \\ &= \frac{10\text{K} + 0.5 \times 10\text{K}}{50\text{K}} \times 100 = \frac{15\text{K}}{50\text{K}} \times 100 \\ &= 30\% \end{aligned}$$

2. Actively searching definition is strict

Applied for a job in the past 4 weeks?

Have you applied for a job in past 12 months?

Do you want a job even if you aren't

"actively searching" in past 4 weeks?

Discouraged workers: initially not in labor force

→ Haven't actively searched in last 4 weeks

→ Searched within the last year.

→ Take a job if offered

UE Rate

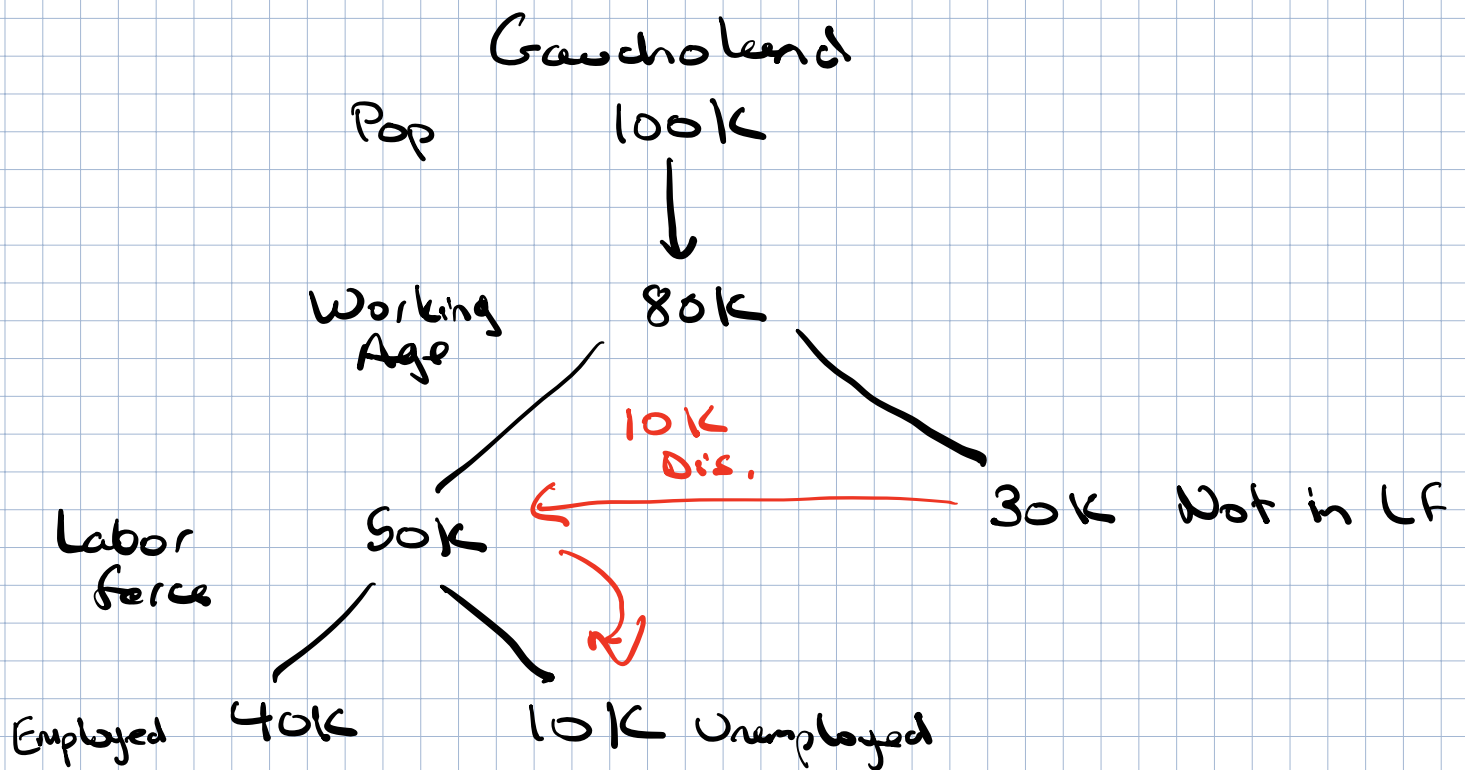
Adjusted for

Discouraged
Workers

$$= \frac{\text{Unemployed} + \text{Discouraged}}{\text{Labor force} + \text{Discouraged}} \times 100$$

$$= \frac{7.5 \text{ mil} + 6.2 \text{ mil}}{171.5 \text{ mil} + 6.2 \text{ mil}} \times 100$$

$$= \frac{13.7 \text{ mil}}{177.7 \text{ mil}} \times 100 = 7.7\%$$



10k Discouraged Workers

$$\text{UE Rate with Discouraged} = \frac{\text{UE} + \text{Dis}}{\text{LF} + \text{Dis}} \times 100$$

$$\begin{aligned} \text{UE Rate} + \text{Dis} &= \frac{10k + 10k}{50k + 10k} \times 100 = \frac{20k}{60k} \times 100 \\ &= 33.3\% \end{aligned}$$